

## JEE Mains 2019 Chapter wise Question Bank

## Communication System - Questions

## Q1

In a communication system operating at wavelength 800 nm, only one percent of source frequency is available as signal bandwidth. The number of channels accommodated for transmitting TV signals of band width 6 MHz are (Take velocity of light  $c = 3 \times 10^8 \text{ m/s}$ ,  $h = 6.6 \times 10^{-34} \text{ J-s}$ )

- (1)  $3.75 \times 10^6$  (2)  $3.86 \times 10^6$   
(3)  $6.25 \times 10^5$  (4)  $4.87 \times 10^5$

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## Q2

A TV transmission tower has a height of 140 m and the height of the receiving antenna is 40 m. What is the maximum distance upto which signals can be broadcasted from this tower in LOS (Line of Sight) mode? (Given : radius of earth =  $6.4 \times 10^6 \text{ m}$ ).

- (1) 65 km (2) 48 km (3) 80 km (4) 40 km

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## Q3

The modulation frequency of an AM radio station is 250 kHz, which is 10% of the carrier wave. If another AM station approaches you for license what broadcast frequency will you allot?

- (1) 2750 kHz (2) 2900 kHz  
(3) 2250 kHz (4) 2000 kHz

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## Q4

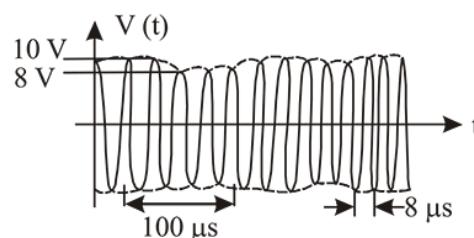
An amplitude modulated signal is given by  $V(t) = 10[1 + 0.3 \cos(2.2 \times 10^4 t)] \sin(5.5 \times 10^5 t)$ . Here  $t$  is in seconds. The sideband frequencies (in kHz) are, [Given  $\pi = 22/7$ ]

- (1) 1785 and 1715 (2) 178.5 and 171.5  
(3) 89.25 and 85.75 (4) 892.5 and 857.5

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## Q5

An amplitude modulated signal is plotted below :



Which one of the following best describes the above signal?

- (1)  $(9 + \sin(2.5 \pi \times 10^5 t)) \sin(2\pi \times 10^4 t) \text{ V}$   
(2)  $(1 + 9 \sin(2\pi \times 10^4 t)) \sin(2.5 \pi \times 10^5 t) \text{ V}$   
(3)  $(9 + \sin(2\pi \times 10^4 t)) \sin(2.5 \pi \times 10^5 t) \text{ V}$   
(4)  $(9 + \sin(4\pi \times 10^4 t)) \sin(5\pi \times 10^5 t) \text{ V}$

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## Q6

A 100 V carrier wave is made to vary between 160 V and 40 V by a modulating signal. What is the modulation index?

- (1) 0.3 (2) 0.5  
(3) 0.6 (4) 0.4

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## Q7

To double the covering range of a TV transmission tower, its height should be multiplied by:

- (1)  $\frac{1}{\sqrt{2}}$  (2) 2  
(3) 4 (4)  $\sqrt{2}$

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## Q8

## Communication System

The wavelength of the carrier waves in a modern optical fiber communication network is close to :

- (1) 2400 nm                      (2) 1500 nm
- (3) 600 nm                      (4) 900 nm

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Q9

In a line of sight radio communication, a distance of about 50 km is kept between the transmitting and receiving antennas. If the height of the receiving antenna is 70m, then the minimum height of the transmitting antenna should be :

(Radius of the Earth =  $6.4 \times 10^6$  m).

- (1) 20 m                      (2) 51 m
- (3) 32 m                      (4) 40 m

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Q10

A signal  $A \cos \omega t$  is transmitted using  $v_0 \sin \omega t$  modulated (AM) signal is:

- (1)  $v_0 \sin \omega_0 t + \frac{A}{2} \sin (\omega_0 - \omega)t + \frac{A}{2} (\omega_0 + \omega)t$
- (2)  $v_0 \sin [\omega_0 (1 + 0.01 A \sin \omega t)t]$
- (3)  $v_0 \sin \omega_0 t + A \cos \omega t$
- (4)  $(v_0 + A) \cos \omega t \sin \omega_0 t$

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Q11

The physical sizes of the transmitter and receiver antenna in a communication system are:

- (1) independent of both carrier and modulation frequency
- (2) inversely proportional to carrier frequency
- (3) inversely proportional to modulation frequency
- (4) proportional to carrier frequency

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Q11

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A message signal of frequency 100 MHz and peak voltage 100 V is used to execute amplitude modulation on a carrier wave of frequency 300 GHz and peak voltage 400 V. The modulation index and difference between the two side band frequencies are :

- (1) 4 ;  $1 \times 10^8$  Hz                      (2) 4 ;  $2 \times 10^8$  Hz
- (3) 0.25 ;  $2 \times 10^8$  Hz                      (4) 0.25 ;  $1 \times 10^{-8}$  T

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Q12

Given below in the left column are different modes of communication using the kinds of waves given in the right column.

- |                                |                   |
|--------------------------------|-------------------|
| A. Optical Fibre Communication | P. Ultrasound     |
| B. Radar                       | Q. Infrared Light |
| C. Sonar                       | R. Microwaves     |
| D. Mobile Phones               | S. Radio Waves    |

From the options given below, find the most appropriate match between entries in the left and the right column.

- (1) A – Q, B – S, C – R, D – P
- (2) A – S, B – Q, C – R, D – P
- (3) A – Q, B – S, C – P, D – R
- (4) A – R, B – P, C – S, D – Q

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Q13

In an amplitude modulator circuit, the carrier wave is given by,  $C(t) = 4 \sin (20000 \pi t)$  while modulating signal is given by,  $m(t) = 2 \sin (2000 \pi t)$ . The values of modulation index and lower side band frequency are :

- (1) 0.5 and 10 kHz                      (2) 0.4 and 10 kHz
- (3) 0.3 and 9 kHz                      (4) 0.5 and 9 kHz

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## Communication System - Answers

Q1

$$(3) \text{ Frequency, } f = \frac{V}{\lambda} = \frac{3 \times 10^8}{8 \times 10^{-7}} = \frac{30}{8} \times 10^{14} \text{ Hz}$$

$$= 3.75 \times 10^{14} \text{ Hz}$$

$$1\% \text{ of } f = 0.0375 \times 10^{14} \text{ Hz}$$

$$= 3.75 \times 10^{12} \text{ Hz} = 3.75 \times 10^6 \text{ MHz}$$

As we know, number of channels accommodated for transmission =

$$\frac{\text{total bandwidth of Channel}}{\text{bandwidth needed per channel}} = \frac{3.75 \times 10^6}{6} = 6.25 \times 10^5$$

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Q2

(1) Maximum distance upto which signal can be broadcasted

$$d_{\max} = \sqrt{2Rh_T} + \sqrt{2Rh_R}$$

where  $h_T$  and  $h_R$  are heights of transmission tower and receiving antenna respectively.

Putting values  $R$ ,  $h_T$  and  $h_R$

$$d_{\max} = \sqrt{2 \times 6.4 \times 10^6} [\sqrt{140} + \sqrt{40}]$$

$$\text{or, } d_{\max} \approx 65 \text{ km}$$

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Q3

(4) According to question, modulation frequency, 250 Hz is 10% of carrier wave

$$f_{\text{carrier}} = \frac{250}{0.1} = 2500 \text{ KHZ}$$

$\therefore$  Range of signal  $2500 \pm 250 \text{ KHZ} = 2250 \text{ Hz to } 2750 \text{ Hz}$  For 2000 KHZ

$$f_{\text{mod}} = 200 \text{ Hz}$$

$\therefore$  Range = 1800 KHZ to 2200 KHZ

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Q4

$$V(t) = 10 [1 + 0.5 \cos(2.2 \times 10^4 t)]$$

$$\sin(5.5 \times 10^5 t)$$

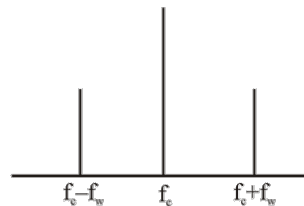
$$= 10 + 1.5 [\sin(57.2 \times 10^4 t) + \sin(52.8 \times 10^4 t)]$$

$$\omega_c + \omega_w = 57.2 \times 10^4 = 2\pi f_1$$

$$f_1 = \frac{57.2 \times 10^4}{2 \times \left(\frac{22}{7}\right)} = 9.1 \times 10^4 \approx 91 \text{ KHz}$$

$$\omega_c - \omega_w = 52.8 \times 10^4$$

$$f_2 = \frac{52.8 \times 10^4}{2 \times \left(\frac{22}{7}\right)} \approx 84 \text{ KHz}$$



Upper side band frequency ( $f_1$ ) is

$$f_1 = f_c + f_w = \frac{52.8 \times 10^4}{2\pi} \approx 85.00 \text{ kHz}$$

Lower side band frequency ( $f_2$ ) is

$$f_2 = f_c - f_w = \frac{57.2 \times 10^4}{2\pi} \approx 90.00 \text{ kHz}$$

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## Communication System

Q5

(3) After analysing the graph we may conclude that

(i) Amplitude varies as  $8-10$  V or  $9 \pm 1$

(ii) Two time period as

$100 \mu\text{s}$  (signal wave) &  $8 \mu\text{s}$  (carrier wave)

So, equation of AM signal is

$$\left[ 9 \pm 1 \sin\left(\frac{2\pi t}{T_1}\right) \sin\left(\frac{2\pi t}{T_2}\right) \right]$$

$$= [9 \pm \sin(2\pi \times 10^4 t)] \sin(2.5\pi \times 10^5 t) \text{V}$$

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Q6

(3) Maximum amplitude  $= E_m + E_c = 160$

$$E_m + 100 = 160$$

$$E_m = 160 - 100 = 60$$

Modulation index,

$$\mu = \frac{E_m}{E_c} = \frac{60}{100}$$

$$\mu = 0.6$$

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Q7

(3) As we know, Range  $= \sqrt{2hR}$

therefore to double the range height 'h' should be 4 times.

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Q8

(2) Carrier waves of wavelength  $1500 \text{ nm}$  is used in modern optical fiber communication.

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Q9

$$(3) \text{ LOS} = \sqrt{2h_T R} + \sqrt{2h_R R}$$

$$\text{or } 50 \times 10^3 = \sqrt{2h_T \times 6.4 \times 10^6} + \sqrt{2 \times 70 \times 6.4 \times 10^6}$$

On solving,  $h_T = 32 \text{ m}$

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Q10

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(1) The equation of amplitude modulated wave

$$m = (v_0 + A \cos \omega t) \sin \omega t$$

$$= v_0 \sin \omega_0 t + A \cos \omega t \sin \omega_0 t$$

$$= v_0 \sin \omega_0 t + \frac{A}{2} [\sin(\omega_0 - \omega)t + \sin(\omega_0 + \omega)t]$$

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Q11

(2) Size of antenna,

$$l = \frac{\lambda}{4}. \text{ As } \lambda = \frac{c}{f} \therefore l \propto \frac{1}{f}$$

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Q11

(3) Range of frequency  $= (f_c - f_m)$  to  $(f_c + f_m)$

$$\therefore \text{Band width} = 2f_m = 2 \times 100 \times 10^6 \text{ Hz}$$

$$= 2 \times 10^8 \text{ Hz}$$

$$\text{and Modulation index} = \frac{A_m}{A_c} = \frac{100}{400} = 0.25$$

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Q12

(3) Optical Fibre Communication – Infrared Light

Radar – Radio Waves

Sonar – Ultrasound

Mobile Phones – Microwaves

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Q13

$$(4) \text{ Modulation index, } \mu = \frac{A_m}{A_c} = \frac{2}{4} = 0.5$$

$$\text{Given, } f_e = \frac{20000\pi}{2\pi} = 10000 \text{ Hz.}$$

$$\text{and } f_m = \frac{2000\pi}{2\pi} = 1000 \text{ Hz.}$$

$$\therefore \text{LSB} = f_e - f_m = 10000 - 1000 = 9000 \text{ Hz.}$$

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